Grade: Check Plus

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ASSIGNMENT 1: ATM

The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	f(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

a. What is the *probability* that a randomly selected customer withdraws *at least $60* from the ATM?

Let X = be the amount withdrawn (in dollars) by a randomly selected customer, with probability distribution f(x) given in the table above. ✓ Qa

"At least $60" - means X ≥ $60

P(X ≥ $60) = 0.15 + 0.25 + 0.05 = **<u>0.45</u>**

0.45 or 45% chance a customer withdraws at least $60 from the ATM.

b. Show by calculation that the *expected value* withdrawn from the ATM is $50 and that the *standard deviation* of the amount withdrawn from this ATM is $26.27. ✓ Qb

To calculate Expected Value E(X) or Mean (μ) = *Sum* of x*f(x):

= 10(0.10) + 20(0.15) + 40(0.30) + 60(0.15) + 80(0.25) + 100(0.05)

= 1 + 3 + 12 + 9 + 20 + 5

= $50

To find Standard Deviation, first will need to find the variance of X, then square root (√) the variance:

Variance(X) = *Sum* of $(x - \mu)^2 f(x)$

$= (10-50)^2(0.10) + (20-50)^2 (0.15) + (40-50)^2 (0.30) + (60-50)^2 (0.15) + (80-50)^2 (0.25) + (100-50)^2 (0.05)$

= 160+135+30+15+225+125

= 690

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Standard Deviation (SD) = √variance
= √690
= $26.27

c. The manager of this convenience score tells you that an expected value of $50 is *not* correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?

✓ Qc

The expected value represents the center of a probability distribution and is calculated as the weighted average of all possible outcomes using their probabilities. Thus, an expected withdrawal of $50 <u>means the long-run average per customer is $50</u>. This does not mean any customer will withdraw exactly $50; individual withdrawals will vary, but over many transactions the average withdrawal will be close to $50.

d. What is the probability that the amount withdrawn from the ATM will fall within *one standard deviation* of the mean?

✓ Qd

P(Mean - SD ≤ X ≤ Mean + SD) =
P(50 - 26.27 ≤ X ≤ 50 + 26.27) =
P(23.73 ≤ X ≤ 76.27)
We take the withdrawal amounts that are between 23.73 and 76.27, which are 40 and 60. Therefore, P = 0.30 + 0.15 = **0.45**

The probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean is 0.45 or 45%.

e. What is the probability that the amount withdrawn from the ATM will fall within *two standard deviations* of the mean?

✓ Qe

P (Mean - 2xSD ≤ X ≤ Mean + 2xSD) =
P (50 - 2x26.27 ≤ X ≤ 50 + 2x26.27) =
P (-2.54 ≤ X ≤ 102.54) - this means that all possible withdrawals (10 to 100) fall in this interval. Therefore, P = **1.00**

1.00, every possible withdrawal lies within two standard deviations of the mean.